Ejercicio 1

Coordenadas naturales

- Se debe elegir un sistema intrínseco a la línea de corriente:

$\hat{es}$: Vector unitario tangente a la línea de corriente ~ dirección del mov.

$\hat{en}$: Vector unitario normal a la línea de corriente, apuntando al centro de curvatura.

Para un flujo estacionario, no viscoso e incompresible, tenemos:

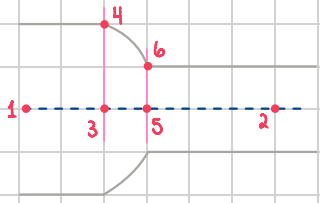
> En dirección tangencial:

$$\frac{\partial V_s}{\partial t} + \frac{\partial}{\partial s}\left(\frac{V^2}{2}\right) = -\frac{1}{\rho}\frac{\partial P}{\partial s} + f_{bs}$$

> En dirección normal:

$$\frac{V^2}{R} = -\frac{1}{\rho}\frac{\partial P}{\partial n} + f_{bn}$$

Comparaciones entre puntos



$$V = \begin{pmatrix} 0 \\ V_s \\ 0 \end{pmatrix}$$

≡ Una línea de corriente

Puntos 1 y 2:

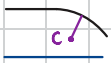
Misma línea de corriente: $P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$

Continuidad:
$Q = V_1 A_1 = V_2 A_2$

$A_1 > A_2$ ... entonces: $V_1 < V_2$ ... entonces: $P_1 > P_2$

· Como la energía cinética aumenta, la presión estática disminuye.

Puntos 3 y 4:



El centro de curvatura apunta hacia la línea central y la presión disminuye a medida que avanzamos en la dirección $\hat{n}$.

$\frac{\partial P}{\partial n} = -\rho \frac{V^2}{R}$ $(\rho > 0,\ V^2 > 0,\ R > 0)$ $\frac{\partial P}{\partial n} < 0 \rightarrow P_4 > P_3$

Puntos 5 y 6:



Cambio de curvatura $\rightarrow P_5 > P_6$

Ejercicio 2

Euler en x, plano horizontal:

$$\frac{\partial u}{\partial t}^{0} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}^{0} + w\frac{\partial u}{\partial z}^{0} = -\frac{1}{\rho}\frac{\partial P}{\partial x}$$

$$\vec{V} = \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} V(x) \\ 0 \\ 0 \end{pmatrix}$$

régimen permanente

$$u\frac{\partial u}{\partial x} = -\frac{1}{\rho}\frac{\partial P}{\partial x}$$

$$(10(1+x)^2)\frac{\partial}{\partial x}(10(1+x)^2) = -\frac{1}{\rho}\frac{\partial P}{\partial x}$$

$$(10(1+x)^2)\cdot 20(1+x) = -\frac{1}{\rho}\frac{\partial P}{\partial x}$$

$$\frac{\partial P}{\partial x} = -200\rho(1+x)^3$$

Ejercicio 3

Vorticidad en z: $\frac{1}{r}\frac{\partial(rV_\theta)}{\partial r} - \frac{1}{r}\frac{\partial V_r}{\partial \theta}$

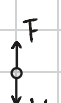
Para $r < r_c$: $\begin{pmatrix}0\\2r\\0\end{pmatrix}$ $\quad \frac{1}{r}\frac{\partial(2r^2)}{\partial r} = \frac{1}{r}\cdot 4r \neq 0$

Para $r > r_c$: $\begin{pmatrix}a/r\\3a/r\\0\end{pmatrix}$ $\quad \frac{1}{r}\frac{\partial(3a)}{\partial r} - \frac{1}{r}\frac{\partial(a/r)}{\partial \theta} = 0$ ✓ Podemos usar Bernoulli con $r \geq r_c$

Puntos: r_2 y L $\qquad \gamma = \rho g$

$$\frac{V_2^2}{2g} + \frac{P_2}{\gamma} = \frac{V_L^2}{2g} + \frac{P_L}{\gamma}$$

$$\frac{10a^2/r_2^2}{2g} + \frac{P_2}{\gamma} = \frac{10a^2/L^2}{2g} + \frac{P_L}{\gamma} \qquad P_L = 5\rho a^2\left(\frac{1}{r^2}+\frac{1}{L^2}\right) + P_2$$



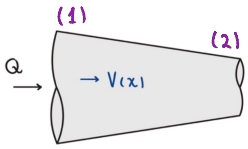
$W = F = P_{(L)}\cdot A$

$$W = \left(5a^2\rho\left(\frac{1}{r^2}+\frac{1}{L^2}\right) + P_2\right)bc$$

Ejercicio 2 - extra

Un flujo permanente, incompresible e inviscido circula por una tubería horizontal de área variable y largo 3 ft. Considere un líquido de densidad $\rho = 103$ kg/m3. La velocidad axial media sobre el eje del conducto está dada por: $V(x) = 10(1 + x)^2$.

Usando la forma diferencial de Euler, determine el gradiente de presión requerido para producir el campo de velocidades. Encontrar dp/dx como función de x.



b) Si la presión en la sección (1) es $P_1 = 50$ psi $(1 \text{ psi} = 6895 \text{ N/m}^2)$, determine la presión en la sección (2)

(1) Integrando el gradiente de presión obtenido en (a) entre x_1 y x_2.

$P_1 = 50 \text{ psi} = 50 \times 6895 = 344{,}75 \text{ KPa}$

> Del desarrollo anterior : $\frac{\partial P}{\partial x} = -200 \rho (1 + x)^3 \quad \left[\frac{Pa}{m}\right]$

Reemplazamos $\rho = 103 \text{ Kg/m}^3$ y dejamos en KPa/m :

$$\frac{\partial P}{\partial x} = -20{,}6 (1 + x)^3$$

> Integramos : $\int_{P_1}^{P_2} dP = \int_{x_1}^{x_2} -20{,}6 (1 + x)^3 \, dx \qquad \begin{cases} x_1 = 0 \\ x_2 = 3 \text{ ft} = 0{,}9144 \text{ m} \end{cases}$

$$P_2 - P_1 = -20{,}6 \left(\frac{1}{4} (1 + x)^4 \right) \Big|_0^{x_2}$$

$$P_2 = P_1 - 5{,}15 ((1 + x_2)^4 - 1)$$

$$P_2 = 280{,}6 \text{ KPa}$$

Fórmulas Certamen 2 :

Euler - Bernoulli - Vorticidad

Ecuación de Euler en coordenadas cartesianas

$$\begin{aligned}
\frac{\partial u_x}{\partial t} + u_x\frac{\partial u_x}{\partial x} + u_y\frac{\partial u_x}{\partial y} + u_z\frac{\partial u_x}{\partial z} &= g_x - \frac{\partial p}{\partial x}\\
\frac{\partial u_y}{\partial t} + u_x\frac{\partial u_y}{\partial x} + u_y\frac{\partial u_y}{\partial y} + u_z\frac{\partial u_y}{\partial z} &= g_y - \frac{\partial p}{\partial y}\\
\frac{\partial u_z}{\partial t} + u_x\frac{\partial u_z}{\partial x} + u_y\frac{\partial u_z}{\partial y} + u_z\frac{\partial u_z}{\partial z} &= g_z - \frac{\partial p}{\partial z}
\end{aligned}$$

Ecuación de Euler en coordenadas cilíndricas

$$\begin{aligned}
\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta} + u_z\frac{\partial u_r}{\partial z} - \frac{u_\theta^2}{r} &= g_r - \frac{1}{\rho}\frac{\partial p}{\partial r}\\
\frac{\partial u_\theta}{\partial t} + u_r\frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta} + u_z\frac{\partial u_\theta}{\partial z} + \frac{u_r\, u_\theta}{r} &= g_\theta - \frac{1}{\rho}\frac{1}{r}\frac{\partial p}{\partial \theta}\\
\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta} + u_z\frac{\partial u_z}{\partial z} &= g_z - \frac{1}{\rho}\frac{\partial p}{\partial z}
\end{aligned}$$

Ecuación de Euler en líneas de corriente, coordenadas (normal, tangencial, binormal)

$$\begin{aligned}
-\frac{\partial v_n}{\partial t} + \frac{v_s^2}{R} &= g_n - \frac{1}{\rho}\frac{\partial p}{\partial n} \qquad (\vec{e}_n)\\
\frac{\partial v_s}{\partial t} + \frac{1}{2}\frac{\partial}{\partial s}(v_s^2) &= g_s - \frac{1}{\rho}\frac{\partial p}{\partial s} \qquad (\vec{e}_s)\\
\frac{\partial v_b}{\partial t} &= g_b - \frac{1}{\rho}\frac{\partial p}{\partial b} \qquad (\vec{e}_b)
\end{aligned}$$

Ecuación de Bernoulli en líneas de corriente con densidad constante

$$\int_C \frac{\partial \vec{u}}{\partial t}\, d\vec{c} + \frac{1}{2}(V_2^2 - V_1^2) + \frac{1}{\rho}\,(p_2 - p_1) + g\,(z_2 - z_1) = 0$$

Vorticidad en coordenadas cartesianas

$$\vec{\omega} = \nabla \times \vec{u} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right)\vec{e}_x + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right)\vec{e}_y + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\vec{e}_z = (\omega_x, \omega_y, \omega_z)$$

Vorticidad en coordenadas cilíndricas

$$\vec{\omega} = \nabla \times \vec{u} = \left(\frac{1}{r}\frac{\partial u_z}{\partial \theta} - \frac{1}{r}\frac{\partial}{\partial z}(r\, u_\theta)\right)\vec{e}_r + \left(\frac{\partial u_r}{\partial z} - \frac{\partial u_z}{\partial r}\right)\vec{e}_\theta + \left(\frac{1}{r}\frac{\partial}{\partial r}(r\, u_\theta) - \frac{1}{r}\frac{\partial u_r}{\partial \theta}\right)\vec{e}_z = (\omega_r, \omega_\theta, \omega_z)$$

Ecuación de Navier Stokes para flujo laminar

$$\frac{\partial \vec{u}}{\partial t} + (\vec{u}\cdot\nabla)\vec{u} = \vec{g} - \frac{1}{\rho}\nabla p + \frac{\mu}{\rho}\nabla^2\vec{u} + \frac{1}{3}\frac{\mu}{\rho}\nabla(\nabla\cdot\vec{u})$$

Flujo incompresible $(\nabla\cdot\vec{u}) = 0 \quad \Longrightarrow \quad \dfrac{\partial \vec{u}}{\partial t} + (\vec{u}\cdot\nabla)\vec{u} = \vec{g} - \dfrac{1}{\rho}\nabla p + \dfrac{\mu}{\rho}\nabla^2\vec{u}$

Navier Stokes flujo incompresible en coordenadas cartesianas

$$\begin{aligned}
\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} &= g_x - \frac{1}{\rho}\frac{\partial p}{\partial x} + \frac{\mu}{\rho}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right)\\
\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} &= g_y - \frac{1}{\rho}\frac{\partial p}{\partial y} + \frac{\mu}{\rho}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right)\\
\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} &= g_z - \frac{1}{\rho}\frac{\partial p}{\partial z} + \frac{\mu}{\rho}\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right)
\end{aligned}$$

Navier Stokes flujo incompresible en coordenadas cilíndricas

$$\begin{aligned}
\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_r}{\partial \theta} + u_z\frac{\partial u_r}{\partial z} - \frac{u_\theta^2}{r} &= g_r - \frac{1}{\rho}\frac{\partial p}{\partial r} + \frac{\mu}{\rho}\left(\frac{\partial^2 u_r}{\partial r^2} + \frac{1}{r}\frac{\partial u_r}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u_r}{\partial \theta^2} + \frac{\partial^2 u_r}{\partial z^2} - \frac{u_r}{r^2} - \frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta}\right)\\
\frac{\partial u_\theta}{\partial t} + u_r\frac{\partial u_\theta}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_\theta}{\partial \theta} + u_z\frac{\partial u_\theta}{\partial z} + \frac{u_r u_\theta}{r} &= g_\theta - \frac{1}{\rho}\frac{\partial p}{\partial \theta} + \frac{\mu}{\rho}\left(\frac{\partial^2 u_\theta}{\partial r^2} + \frac{1}{r}\frac{\partial u_\theta}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u_\theta}{\partial \theta^2} + \frac{\partial^2 u_\theta}{\partial z^2} + \frac{2}{r^2}\frac{\partial u_r}{\partial \theta} - \frac{u_\theta}{r^2}\right)\\
\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + \frac{u_\theta}{r}\frac{\partial u_z}{\partial \theta} + u_z\frac{\partial u_z}{\partial z} &= g_z - \frac{1}{\rho}\frac{\partial p}{\partial z} + \frac{\mu}{\rho}\left(\frac{\partial^2 u_z}{\partial r^2} + \frac{1}{r}\frac{\partial u_z}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2}\right)
\end{aligned}$$